Mathematics Formulae Sheet

Quick revision for JAMB UTME

Indices and logarithms

- $a^m \times a^n = a^{m+n}$; $a^m \div a^n = a^{m-n}$; $(a^m)^n = a^{mn}$
- $a^0 = 1$; $a^{-n} = 1/a^n$; $a^{1/n}$ = n-th root of a
- If $a^x = N$ then $\log_a N = x$. $\log(AB) = \log A + \log B$; $\log(A/B) = \log A - \log B$; $\log A^n = n \log A$

Algebra

- Quadratic $ax^2 + bx + c = 0$: $x = [-b \pm \sqrt{(b^2 - 4ac)}] / 2a$. Sum of roots = $-b/a$; product = c/a
- Difference of two squares: $a^2 - b^2 = (a - b)(a + b)$
- Arithmetic progression: $T_n = a + (n - 1)d$; $S_n = n/2\,[2a + (n - 1)d]$
- Geometric progression: $T_n = ar^{n-1}$; $S_n = a(r^n - 1)/(r - 1)$; $S_\infty = a/(1 - r)$ for $|r| < 1$
- Variation: direct $y = kx$; inverse $y = k/x$; joint $y = kxz$

Sets and probability

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- Probability = number of favourable outcomes ÷ total outcomes. P(A or B) = P(A) + P(B) for mutually exclusive events; P(A and B) = P(A) × P(B) for independent events
- Permutations: nPr = n!/(n − r)!. Combinations: nCr = n!/[r!(n − r)!]

Geometry and mensuration

Shape	Area / Volume / Surface
Triangle	½ × base × height; or ½ab sin C
Circle	Area πr^2; circumference $2\pi r$
Sector	Area $(\theta/360) \times \pi r^2$; arc length $(\theta/360) \times 2\pi r$
Trapezium	½(a + b)h
Cylinder	Volume $\pi r^2 h$; curved surface $2\pi rh$
Cone	Volume $\frac{1}{3}\pi r^2 h$; curved surface πrl
Sphere	Volume $\frac{4}{3}\pi r^3$; surface $4\pi r^2$

- Pythagoras: $a^2 + b^2 = c^2$. Interior angles of an n-sided polygon sum to $(n - 2) \times 180°$

- Angle at the centre is twice the angle at the circumference on the same arc; angles in the same segment are equal; opposite angles of a cyclic quadrilateral add to 180°

Trigonometry

- sin θ = opp/hyp; cos θ = adj/hyp; tan θ = opp/adj = sin θ / cos θ. $\sin^2\theta + \cos^2\theta = 1$
- Sine rule: $a/\sin A = b/\sin B = c/\sin C$. Cosine rule: $a^2 = b^2 + c^2 - 2bc\cos A$
- sin 30° = ½; cos 60° = ½; sin 45° = cos 45° = 1/√2; tan 45° = 1; sin 60° = cos 30° = √3/2; tan 30° = 1/√3; tan 60° = √3

Coordinate geometry

- Distance between (x_1, y_1) and $(x_2, y_2) = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$; midpoint = $((x_1 + x_2)/2, (y_1 + y_2)/2)$
- Gradient $m = (y_2 - y_1)/(x_2 - x_1)$. Line: $y = mx + c$. Parallel lines have equal gradients; perpendicular gradients multiply to −1

Statistics

- Mean = Σfx/Σf. Median = middle value when ordered. Mode = most frequent value. Range = highest − lowest
- Standard deviation = $\sqrt{[\Sigma(x - \text{mean})^2 / n]}$

Calculus

- $d/dx\,(x^n) = nx^{n-1}$; $\int x^n\,dx = x^{n+1}/(n + 1) + C$ $(n \neq -1)$
- $d/dx\,(\sin x) = \cos x$; $d/dx\,(\cos x) = -\sin x$. Turning points where $dy/dx = 0$

Exam tip: Write the formula first, then substitute. It earns method marks and prevents slips.